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Terang
Energy Access Beyond the Grid

Hackathon Proposal

Executive Summary

While Indonesia reports an official village electrification ratio of 99.90%, national utility figures reveal that only 92.75% of villages are connected to the grid. The remaining gap is filled by unmonitored solar lanterns or small diesel systems that quickly fail, leaving over 4,808 unconnected villages - concentrated primarily across eastern Indonesia and archipelagic ASEAN - invisible to planners, utilities, and investors. While macro-initiatives like ASEAN Power Grid (APG) focus on interconnecting national grids across mainlands, Terang addresses the millions of people in island communities that exist beyond the physical and economic reach of any grid.

We have tested this approach in Maluku province, analysing all 1,233 villages using free satellite nighttime-light imagery. By replacing standard built-up pixel masking with an unmasked polygon-based reduction - recovering the 79% of remote settlements typically lost in standard geospatial analyses - we identified 111 settled villages emitting light at or below uninhabited baseline levels, alongside 894 villages suffering from severe, seasonal service intermittency (Figure 1).

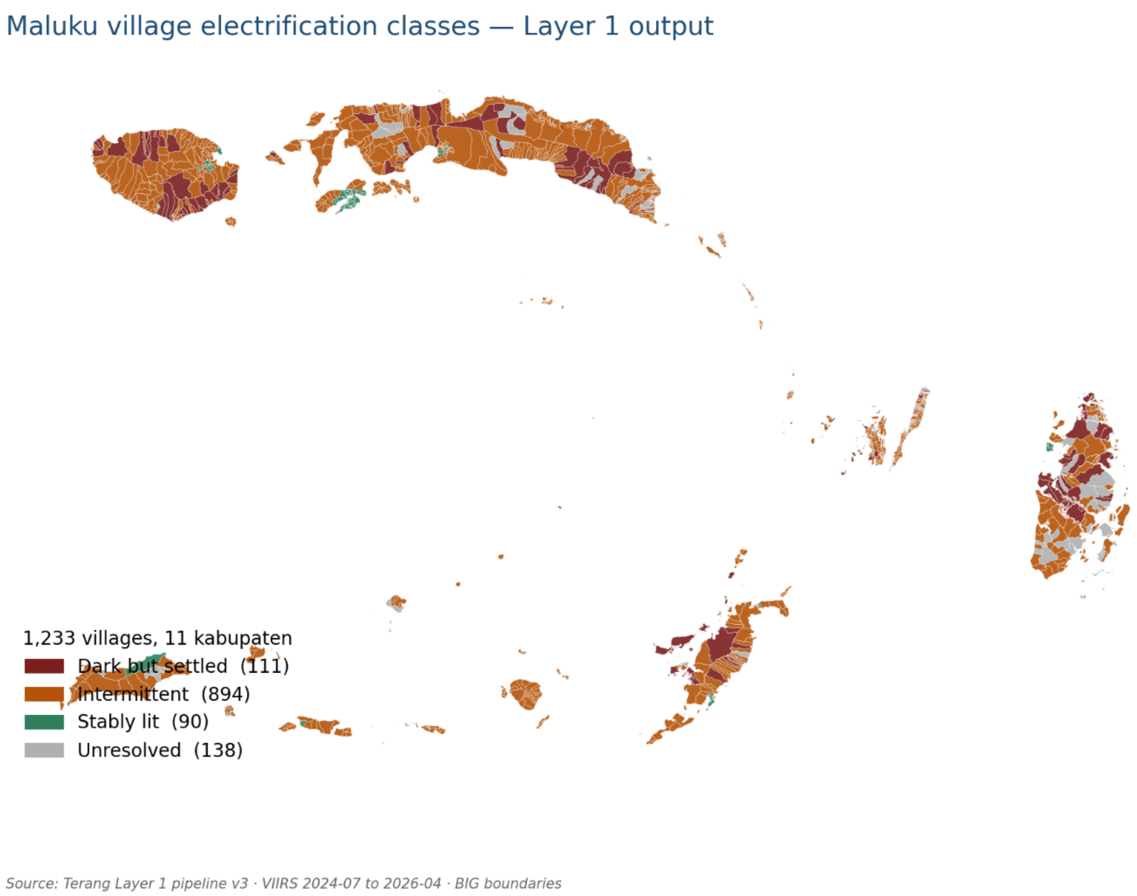

Figure 1. Maluku province gap map. All 1,233 villages coloured by detection class – dark but settled (red), intermittent (amber), stably lit (green), unresolved (grey outline).

Terang translates these satellite observations into an end-to-end, investable off-grid pipeline through a three-layer architecture:

- **Layer 1 (Detect)**: Uses VIIRS nighttime-light data and settlement footprints to pinpoint dark and intermittent villages, offering cheap, continuous, independent observation where official records fail.
- **Layer 2 (Prescribe)**: Applies an auditable, rule-based decision tree that screens sites for environmental and policy constraints (directing protected-area overlaps into policy pathways and grid-proximate sites to utility extension) before matching remaining dark villages to the least-cost technology indicated by local geography (e.g., run-of-river micro-hydro, solar-biomass, or solar-battery systems).
- **Layer 3 (Pool):** Bundles individual village mini-grids into district-level portfolios for institutional blended finance based on four criteria: geographic proximity, technology homogeneity, beneficiary profile, and intervention model. Each candidate bundle is

then validated through a financial, socioeconomic, and environmental benefit analysis quantifying each economical value of the project, income generation, employment, and carbon offsets, translating technical screening into investment opportunities that minimising the risk of capital deployment beyond the grid.

Beyond technical verification, Terang operates as a sustainable social enterprise. By leveraging open, global datasets – requiring only country-specific boundary files – the framework is scalable to comparable island geographies across ASEAN, with the Philippines as a natural second deployment target. Terang verification model costs a fraction of conventional field audits' price, freeing the difference for reinvestment into capital funding, system maintenance, or community-level reliability measures. Ultimately, Terang bridges the gap between capital and off-grid communities, ensuring that energy investments yield verifiable, long-term socioeconomic and environmental resilience.

Introduction

Background

The Directorate General of Electricity reports a village electrification ratio in Indonesia of 99.90% (Figure 2), while the same figures show only 92.75% of villages served by PLN, the national utility (ESDM-Electrification Statistics, 2023; PLN Annual Report, 2023). The gap between those numbers is where this proposal begins. A village counted as electrified may have a 24-hour grid connection, or a batch of solar lanterns distributed five years ago. Both are recorded identically, and when lantern or diesel systems fail within two to three years, no institutional record captures the reversal.
The shortfall is concentrated in eastern Indonesia: 4,808 villages have no PLN connection, 4,398 of them in Papua and the rest across Maluku, Nusa Tenggara and Sulawesi, the small, dispersed, maritime communities, expensive to survey and easy to overlook (ESDM-Electrification Statistics, 2023; PLN Annual Report, 2023). Because the statistics say the problem is nearly solved, no data trail identifies which villages remain unserved. Without that list, budgets are allocated on aggregates and investors cannot readily identify projects. We have selected Maluku as the pilot province rather than Papua because its boundary data is complete, its scale permits an end-

to-end run within the designated timeframe of analysis, and its archipelagic conditions are broadly representative of the larger gap in Papua and of comparable island geographies elsewhere in ASEAN.

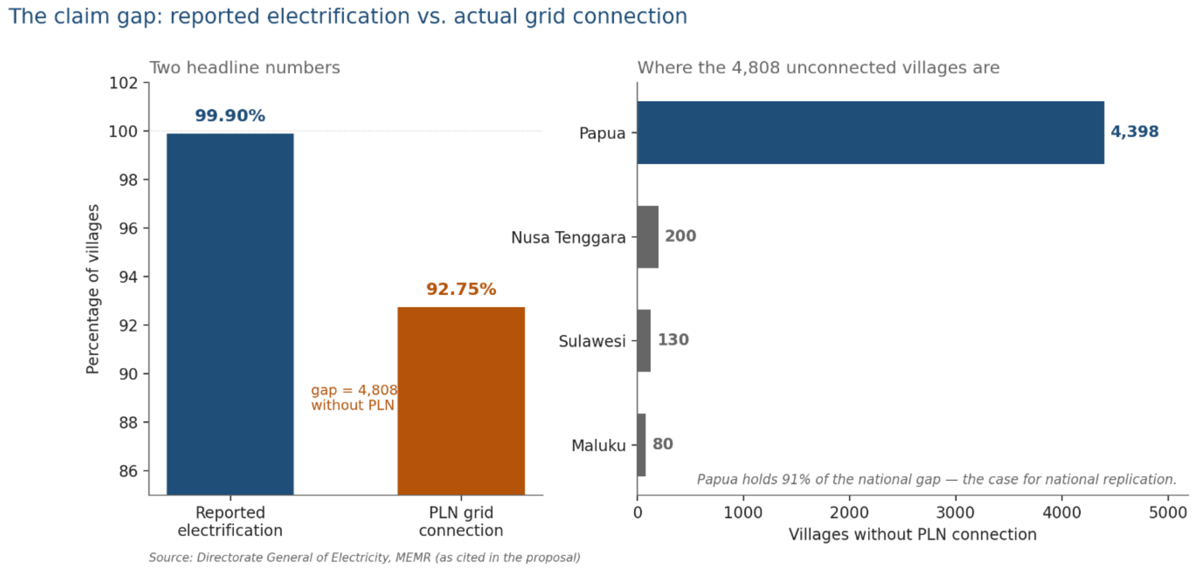

Figure 2. Reported village electrification ratio (99.90%) against villages served by PLN (92.75%), with the 4,808 unconnected villages disaggregated by province.

Relevance

ASEAN is preparing to invest in cross-border transmission through the ASEAN Power Grid (APG) and recent analyses by the World Bank highlight that this project will need approximately $800 billion USD (Centre for Strategic and International Studies, 2026). However, ASEAN's energy challenge extends far beyond connecting the mainlands - the region is home to vast archipelagic nations like Indonesia and the Philippines, where the main reason remote communities lack electricity is not neglect, but brutal terrain, prohibitive grid extension costs, or unaffordable standard tariffs. Min et al. (2024) define energy poverty as the condition where a settlement's electricity consumption is so low or so intermittent that it is statistically indistinguishable from having no supply at all. Within countries, most variation in energy poverty is explained by precisely these local factors — population density, terrain, and proximity to existing infrastructure — rather than by national income or policy, which means that village-level detection, not provincial aggregates, is the appropriate resolution for intervention.

This is where Terang addresses energy security and accessibility beyond the grid. Highlighted at SAYEF 2026, Terang locates communities that current systems cannot see – pinpointing named villages with exact coordinates and custom recommendations. On energy security, it advocates for distributed generation in remote islands to replace price-volatile, storm-vulnerable diesel shipments with resilient local power, utilizing satellite monitoring for real-time early warning when outages occur.

While the APG program connects national grids to one another, Terang addresses the communities beyond every grid. Because its core methodology relies on free, global datasets, requiring only country-specific village boundary files, it is readily scalable across the region, making island-heavy nations like the Philippines a natural candidate for the next deployment.

Objectives

1. Detect. Identify villages where official claims and satellite observation disagree, at village level rather than in aggregate, using only free data.

2. Prescribe. Match each flagged village to the least-cost technology its geography appears to support, after separating sites that fall within legally protected areas.

3. Pool. Aggregate individual village projects into district portfolios large enough for institutional capital.

4. Verify. Replace slow, costly field verification with continuous satellite monitoring, so milestones can be confirmed cheaply and repeatedly.

5. Sustain. Use continuous satellite observation to ensure electrification persists after installation.

6. Quantify. Express project value not only in financial returns but in a structured socioeconomic and environmental benefit framework, covering employment, household welfare, emission reduction, and electrification reliability, so that concessional capital can be justified on development grounds alongside commercial ones.

7. Replicability. Publish the methodology openly, for reuse across ASEAN.

Problem Statement

Rural electrification in archipelagic Southeast Asia fails at three points, and each failure compounds the next.

Measurement. Statistics record connections, not service. A village with three broken lanterns and a village with 24-hour grid power appear identical. Because that choice determines where budgets go, the communities most in need are hidden by the figures meant to describe them. Satellite analysis across 115 developing countries finds 60 per cent more people living in energy poverty than official electricity-access counts recognise, confirming that the gap between recorded connections and actual service is structural, not anecdotal (Min et al., 2024).

Origination. A single village mini grid may cost 200,000 US dollars (ESMAP, 2019), far below the minimum ticket for institutional investors, yet above what any individual community can self-finance. From the investor side, facilities mandated to fund energy access consistently report difficulty finding verified bankable projects. From the regulator side, electricity is a basic need under Indonesian law, but without a reliable village-level inventory of who is served. Budget allocation defaults to provincial aggregates rather than targeting the communities that remains without power. Capital and need coexist with no mechanism or list to connect them.

Persistence. Finance pays on installation. Once a system is commissioned nobody is paid to confirm it still works, and field verification consumes 10 to 15 per cent of programme costs, so it is done once and rarely repeated. The result is systems built, counted, then quietly failing. Terang addresses all three with one capability: cheap, continuous, independent observation.

Proposed Solution

Solution Concept and Description

In plain terms: we use satellites to see which villages are actually dark at night, work out what should be built in each, and bundle them into investment packages validated by financial, socioeconomic, and environmental benefit analysis. Three layers, each feeding the next as shown in Figure 3.

Layer 1 – Detect. Satellites operated by NASA and NOAA record light emitted at night. We compare this against settlement footprints for every village. Villages that are demonstrably inhabited yet emit light at or below the radiance of uninhabited terrain are flagged for attention. Cross-referencing against official electrification records – which would allow flagged villages to be described as claimed-electrified rather than simply dark – is the next data step and depends on a village-code crosswalk currently under development. Output: a ranked, confidence-scored map of the apparent access gap. The HREA project has validated settlement-level nighttime-light classification against household surveys across 115 countries with correlation coefficients of 0.82 to 0.89 (Min et al., 2024); Terang adapts the principle to administrative village polygons, the unit through which Indonesian budgets and programmes are allocated.

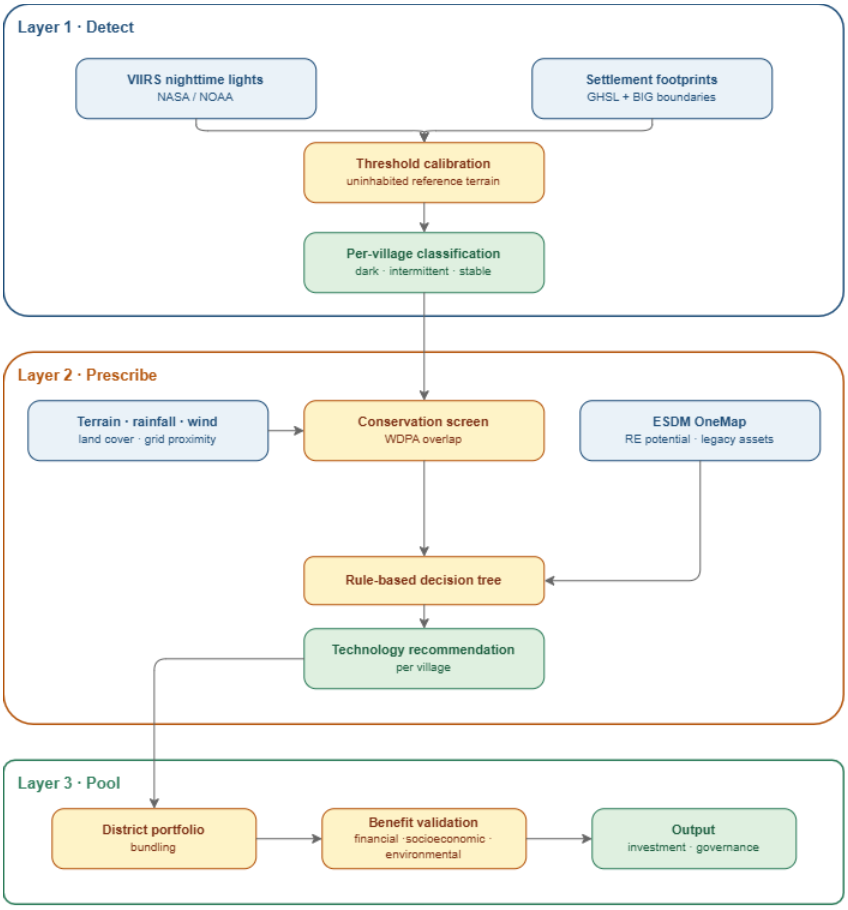

__Figure 3.__ Terang three-layer architecture: Detect, Prescribe, Pool, showing the data inputs to each layer and the outputs passed downstream. The verification and disbursement stages shown in the diagram are described under Future Possibilities as the next operational extension.

Layer 2 – Prescribe. A transparent decision tree recommends the least-cost technology indicated by terrain, river catchment, rainfall, wind, land cover and distance to the nearest available electrical grid. The tree evaluates two contextual conditions before any commercial-viability logic. Sites where more than ten per cent of the village area falls within a legally protected area are separated from the commercial pipeline and directed toward a policy pathway: rather than proposing infrastructure inside a national park, these villages are flagged for coordinated engagement between the Ministry of Environment and Forestry, the relevant park authority and the local community, in a process that may take years and that Terang does not itself undertake. Sites within approximately five kilometres of a reliably lit settlement (ESMAP, 2020; Mentis et al., 2017) are separated from the off-grid pipeline and directed to grid extension, which under Indonesian regulation is PLN's responsibility rather than a private-developer opportunity; Terang's role for these villages is to hand the identified list to PLN and to the regional government, not to design their electrification. Only after these two contextual filters does the tree evaluate hydro potential (IEA-Hydropower Special Market Report, 2021; ESMAP, 2018), wind speed (IRENA-Future of Wind, 2019; Global Wind Atlas 3.0-DTU/World Bank, 2021), biomass feedstock and the solar-battery default. To refine this resource screening, we integrate localized geospatial data from Indonesia Ministry of Energy and Mineral Resources (ESDM)'s Geoportal OneMap - mapping (ESDM One Map, 2024), both regional renewable energy potential and existing onshore conventional energy infrastructure. Incorporating conventional assets allows the model to identify opportunities for asset repurposing or hybrid configurations (such as solar-diesel systems), optimizing existing legacy generation while transitioning remote communities toward reliable, low-carbon power. Extending the conservation screen to intact forest that is present but not formally protected is the next methodological step and will require land-cover time-series analysis. Output: a recommendation per village with indicative size, cost band, and the decision rule that produced it.

Layer 3 - Pool. Villages are bundled into district portfolios four weighted axes: technology homogeneity, beneficiary profile, resource similarity, and intervention type, with geographic proximity handled by Kecamatan (sub-district) scoping. This bundling is done to squarely aggregating the project. This is to reduce aggregate transaction costs and makes small, fragmented projects attractive to financiers who would otherwise bear high due-diligence costs per project relative to deal size (Covenant of Mayors for Climate & Energy Office, 2023). Each candidate bundle is then evaluated through financial analysis, socioeconomic cost-benefit analysis, and environmental benefit analysis covering the economic value of the project, income generation, employment, and carbon offsets. Bundles that clear validation is structured as a single district portfolio vehicle drawing capital from concessional sources (such as ASEAN Catalytic Green Finance and Multilateral Development Banks), commercial impact debt, and a guarantee providing partial risk cover. Bundles that demonstrate socio-environmental impact but lack sufficient economic return are routed to purely concessional or grant funded instruments. After deployment, service persistence is monitored through the same VIIRS radiance pipeline used for initial detection, flagging degradation for intervention without requiring field visits.

Terang positions itself as a sustainable enterprise designed to scale across diverse island geographies, means, by leveraging open, global datasets, the framework can be rapidly deployed in comparable contexts throughout ASEAN, with the Philippines emerging as a natural target. Terang also functions as a monitoring tool that transforms a rural electrification project inro verifiable, investable portfolios. In doing so, it ensures that every dollar of investment (whether from investors or the government itself), translates into measurable resilience, strengthening the livelihood, promoting green source of energy, and embedding reliability for communities that would otherwise remain invisible.

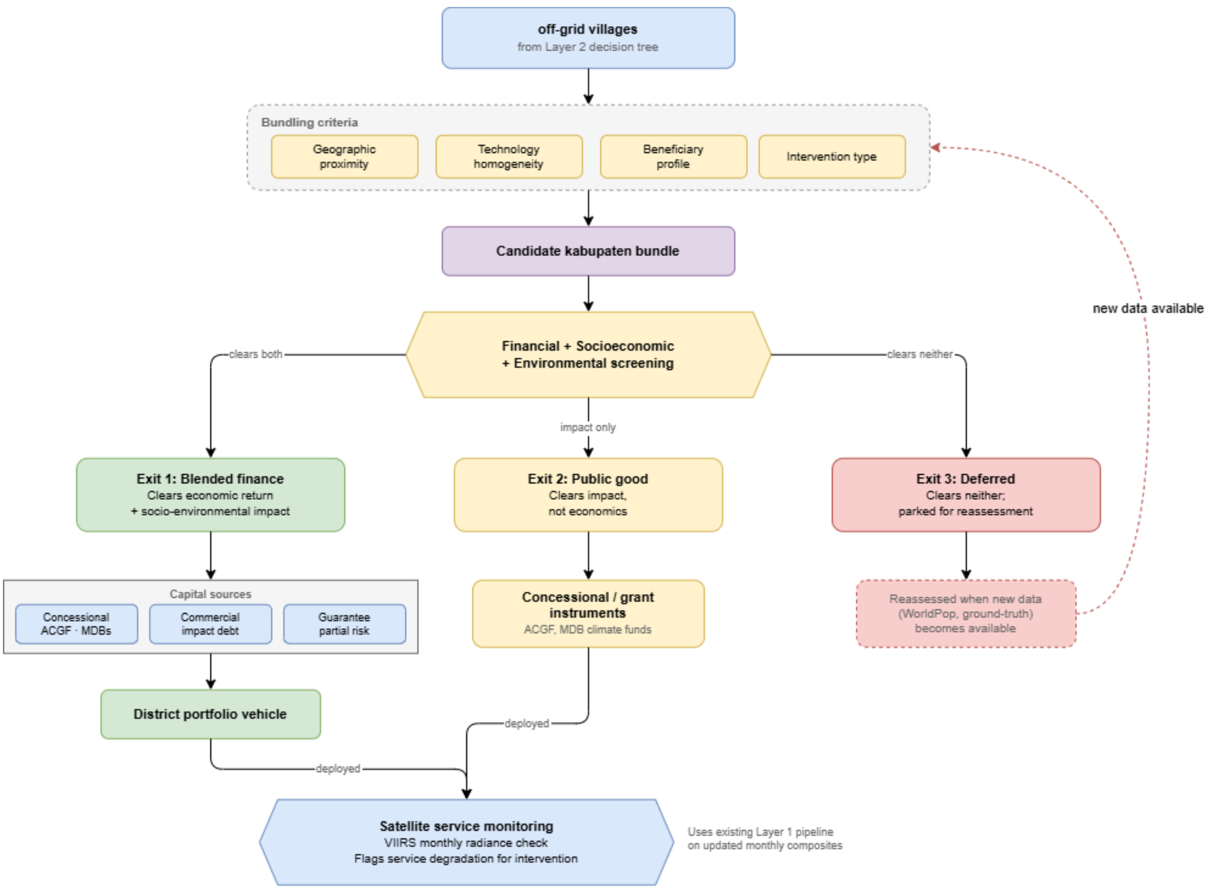

Figure 4. *Layer 3 portfolio structure. Villages are grouped by four weighted axes into candidate bundles, then screened on financial, socioeconomic, and environmental criteria. Bundles that clear both economic and impact*

thresholds enter a blended finance vehicle. Bundles that clear impact only is routed to concessional or grant instruments. Bundles that clear neither are deferred for reassessment.

Scope of Work

Operational prototype. Layer 1 has been built and run on real data for Maluku: 1,233 villages across 11 Regency, 22 months of imagery, with a detection threshold calibrated from uninhabited reference terrain. Layer 2 has been built end-to-end across the 1,005 flagged villages, incorporating resource screening and protected-area screening against the World Database on Protected Areas. A working demonstration application presents all three layers. Our Layer 1 detection closely aligns with several recent ground reports indicating around 130 unelectrified villages in Maluku (bedahusantara.com, 2025) (matching our 111 dark villages classifications), and ESDM allocation plan for the 2026 Village Electrification program across 1,520 locations (Azura-bloombergtechnoz.com, 2026).

In progress and future extension. Screening for intact forest using Hansen land-cover time series, which our present implementation does not yet compute reliably and which we therefore treat as unresolved; the village-code crosswalk that would link radiance measurements to official electrification claims; and deployment to a second province with a larger recorded access gap.

Importantly, our Layer 2 technology prescription pipeline is undergoing active expansion to incorporate a broader array of localized spatial datasets. While baseline resource screening for hydro, wind, biomass, and solar potential is functional, we are currently integrating high-resolution geospatial layers from ESDM's Geoportal OneMap. This includes incorporating regional renewable energy resource mapping and onshore conventional infrastructure data, such as existing diesel assets (PLTD) to evaluate asset repurposing and hybrid configurations like solar-diesel systems. Because these multi-source data streams are still being harmonized, the final decision-tree thresholds and technology mixes remain subject to further calibration.

Finally, the analytical outputs of Layer 2 are being connected to the application's decision-support engine (Layer 3) to run the bundling process up until full economic, SCBA, and environmental benefit analyses. Monetized metrics covering project economics parameters (capital/investment assumptions, cost, and revenue stream), tariffs, household fuel expenditure savings, productive-use income uplift, local job creation, and carbon offset accounting are currently being coded into the application's backend. Completing these modules will allow the system to automatically generate de-risked impact profiles per village cluster, preparing the platform for full provincial validation in Maluku before expanding to a second pilot province with a larger recorded access gap, such as Papua.

Out of scope. Terang does not build, own or operate generation assets, and does not replace engineering feasibility studies, hydrological survey or advanced environmental impact assessment. Its outputs are screening-level instruments directing where more expensive investigation should go.

Methodology

Detection. VIIRS is a sensor on polar-orbiting weather satellites measuring light emitted at night. We use 22 monthly composites at roughly 500 metre resolution, with the Global Human Settlement Layer for built-up area and village boundaries from Indonesia's national geospatial agency. The threshold separating lit from unlit is not taken from literature but calibrated per study area, by sampling radiance at points at least two kilometres from any built-up surface and inland from water. For Maluku, this procedure yielded a background floor of 0.201 nW/cm²/sr and a detection threshold of 0.292 nW/cm²/sr. Reference points were additionally masked against inland water using the JRC Global Surface Water dataset, so that the threshold reflects genuinely uninhabited terrestrial radiance rather than a mixed land-water noise floor.

A methodological correction. Standard practice masks radiance to built-up areas before analysis. At VIIRS resolution this eliminated 79 per cent of Maluku's villages, because settlements smaller than one 500 metre pixel lose every pixel to the mask – and those excluded were systematically the smallest and most remote, precisely the population the project exists to find. We corrected it by treating the village polygon as the unit of analysis, keeping settlement data as a size control (Figure 5).

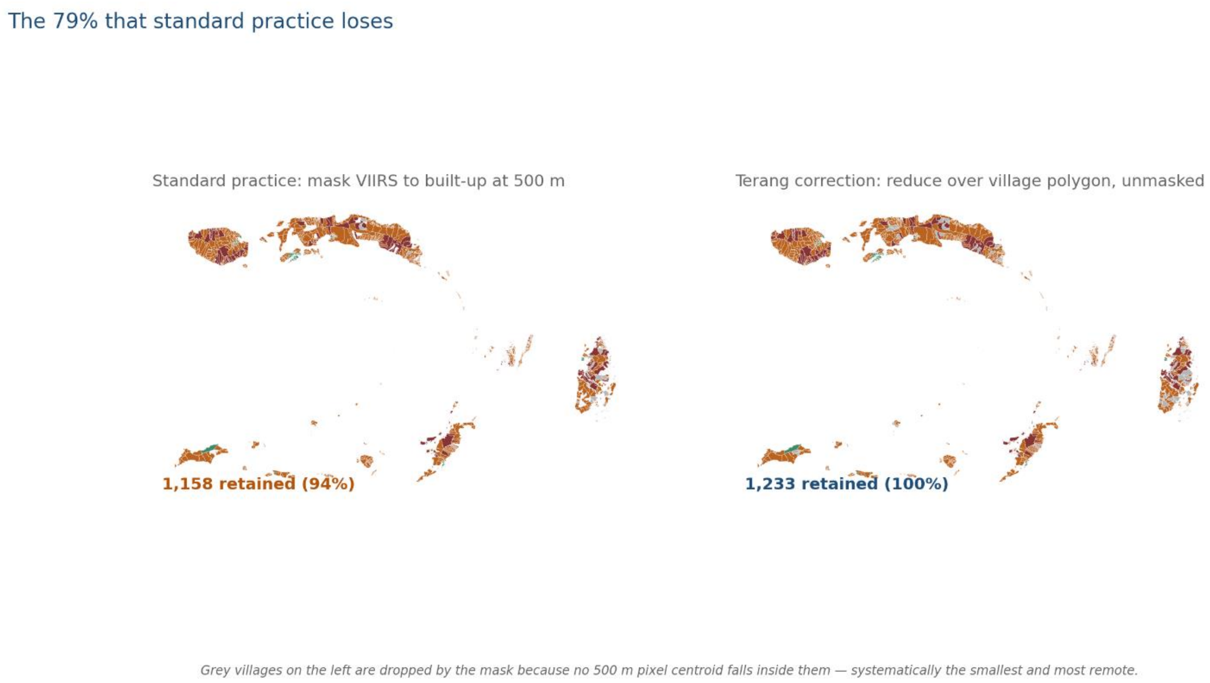

Figure 5. Effect of the masking correction. Left: villages retained under standard built-up masking at VIIRS resolution. Right: villages retained under polygon-based reduction. 79 per cent of villages, systematically the smallest and most remote, are recovered.

Classification and validation. Villages are grouped according to how frequently they emit light above the threshold and how variable that emission is: steady illumination is consistent with grid-quality service, intermittent emission with diesel or degrading solar systems, and absence of emission in a settled area with an access gap. Applied to Maluku, this produced 111 villages classified as dark but settled, 90 as stably lit, and 138 as too small or too frequently cloud-obscured for confident classification, with the remainder classified as intermittent. Within that intermittent group, the proportion of villages emitting light above threshold varies substantially by month and falls sharply during the wet season, a pattern we consider more consistent with genuine service degradation than with measurement noise, though distinguishing the two conclusively would require ground data.

Two checks were applied. The eight brightest villages identified are all located in Ambon city, the provincial capital, at 14 to 29 $nW/cm^2/sr$ against a provincial median of 0.251 $nW/cm^2/sr$, which is consistent with expectation. Separately, the commissioning detector (the same function proposed for milestone verification under Future Possibilities) independently identified January 2017 as the month in which Labuang village in Buru Selatan was electrified, from radiance alone and without ground reporting. That village's light output has visibly degraded since 2022, which we take as indicative of the persistence problem this project seeks to address (Figure 6).

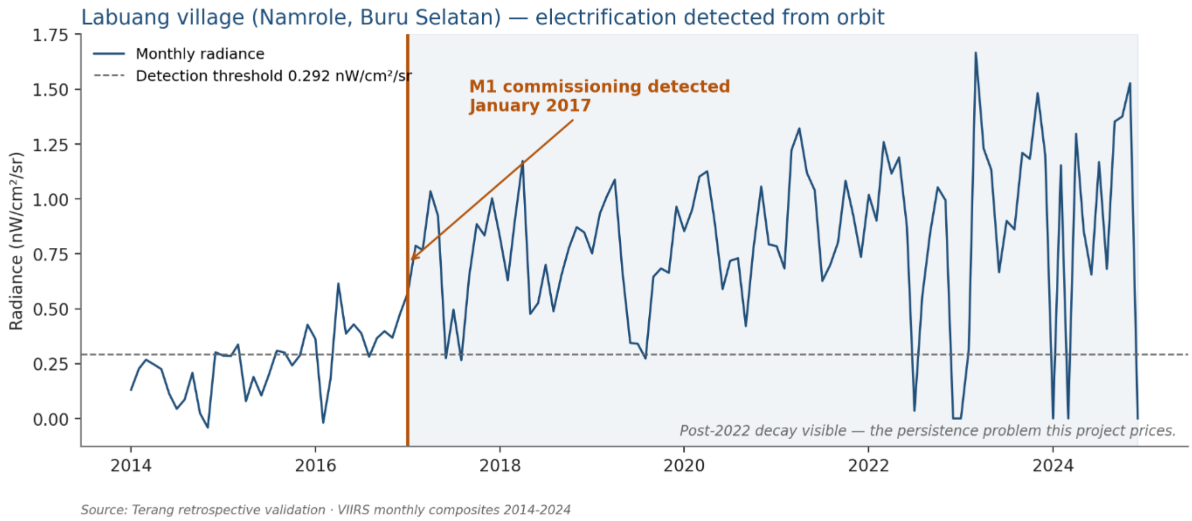

Figure 6. Labuang village, Namrole, Buru Selatan. Monthly radiance 2014-present against the 0.292 nW/cm²/sr detection threshold, with the M1 commissioning step automatically detected at January 2017 and the post-2022 degradation visible.

Related published work, notably the High Resolution Electricity Access project (University of Michigan, with the World Bank and NOAA), offers independent support for the proposition that VIIRS radiance tracks settlement-level electrification (Min, B., et al., 2024). Terang seeks to extend that body of work operationally, translating per-village detection into a technology prescription and a validated, bankable project pipeline.

Prescription. Screening combines elevation and slope, river catchment, rainfall, wind, land cover and distance to the nearest reliably lit settlement. A rule-based decision tree as depicted in Figure 7 assigns technology rather than a statistical model, because a published tree with visible thresholds can be audited and corrected by domain experts. Applied to Maluku's 111 dark villages (Figure 8), 24 were found to fall within protected areas above a ten per cent overlap threshold and were separated from the commercial pipeline for policy engagement, and 14 were directed to grid extension as PLN's statutory responsibility. The remaining 73 villages form the tractable off-grid pipeline: 32 micro-hydro candidates, 31 suited to solar-biomass hybrids, 8 solar-battery and 2 wind-solar. We do not assert a ranking within any single technology tier: the highest-scoring micro-hydro sites all saturate the conservative catchment and head caps we apply, and differentiating between them would require field survey.

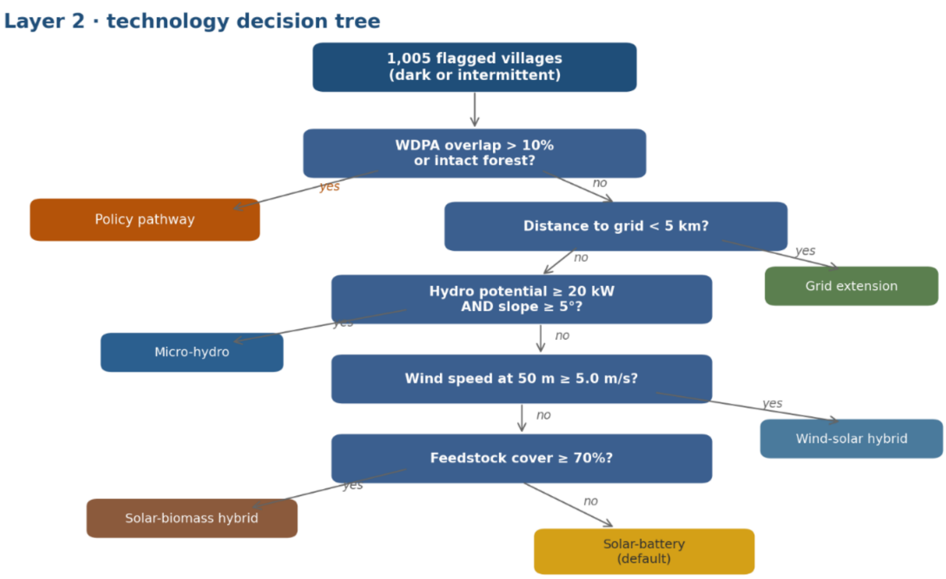

Figure 7. Layer 2 technology decision tree, showing every branch and threshold in sequence: conservation screen, grid proximity, micro-hydro potential, wind speed, biomass feedstock, solar-battery default.

The decision tree in Figure 7 operates as a sequential filter. The first gate is the **conservation screen**: any village whose boundary overlaps a gazetted protected area (WDPA) by more than ten per cent is removed from the commercial pipeline and routed to a policy pathway – a coordinated engagement process between the Ministry of Environment and Forestry, the relevant park authority, and the local community that Terang flags but does not itself undertake. The second gate is **grid proximity**: villages within approximately five kilometres of a reliably lit settlement are directed to PLN for grid extension, since under Indonesian regulation this is the utility's statutory obligation, not a private-developer opportunity.

Only villages that clear both gates enter the technology assignment sequence. The tree first tests for micro-hydro viability: where effective catchment area, elevation head, and rainfall combine to yield a run-of-river potential above the minimum threshold, micro-hydro is assigned as the least-cost option. If hydro conditions are not met, the tree evaluates wind speed at 50 metres; sites meeting the wind threshold receive a wind-solar hybrid prescription. Next, biomass feedstock availability (measured by agricultural land-cover fraction) is tested; qualifying villages receive a solar-biomass hybrid prescription. All remaining villages default to solar-battery systems, the technology with the widest applicability but typically the highest per-kWh cost. At every branch, the threshold values and the decision logic are published and auditable — no assignment is a black box.

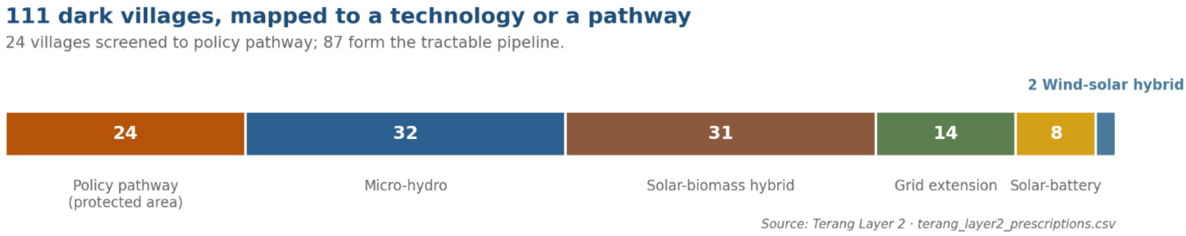

Figure 8. Technology prescription mix for Maluku's 111 dark villages: 24 directed to policy pathway (protected area), 14 to grid extension (PLN mandate); the remaining 73 villages form the tractable off-grid pipeline, comprising 32 micro-hydro, 31 solar-biomass hybrid, 8 solar-battery and 2 wind-solar candidates.

Figure 8 quantifies the output of this decision tree for Maluku's 111 dark-settled villages. The 24 villages directed to the policy pathway are not abandoned; they represent communities where electrification requires navigating overlapping jurisdictions between energy mandates and conservation law. For these villages, Terang's contribution is to make them visible and named, to provide the satellite-derived evidence of their energy status, and to hand that evidence package to the institutions with the authority to act. The 14 villages directed to grid extension similarly require no technology prescription from Terang: they are close enough to existing supply that PLN's mandate applies, and the pipeline's role is to ensure PLN and district governments receive a verified, prioritised list rather than an undifferentiated backlog.

The remaining **73 villages** form the tractable off-grid pipeline. The technology mix reflects Maluku's physical geography: **micro-hydro** dominates in the high-rainfall uplands of Seram and Buru, where steep terrain and reliable precipitation create the conditions for run-of-river systems. **Solar-biomass hybrids** concentrate in the sago and coconut belt, where agricultural residue provides a dispatchable complement to photovoltaic generation. **Solar-battery** systems serve villages in flatter terrain without biomass or hydro resources. The two **wind-solar hybrid** candidates occupy exposed coastal locations where wind speeds at 50 metres exceed the viability threshold. Cross-validation against Kementerian ESDM's official resource registry confirms 98% of micro-hydro and 98% of solar prescriptions, providing independent confidence that the satellite-derived recommendations align with government-mapped resource potential.

Screening also rules options out. Tidal and marine energy, superficially attractive for an island province, remain pre-commercial and unfinanceable at village scale. Solar irradiance across equatorial Indonesia is so uniform it cannot discriminate between sites, so it informs sizing but never site selection.

Limitations. Nighttime lights measure outdoor and aggregate illumination, so an electrified village with very low night-time consumption can appear dark. Radiance confirms village-level energisation and persistence; it does not confirm household connection counts, tariff payment or emissions quantification, each of which needs metered or field data. Terang is a screening instrument that directs ground-truthing, never an audit verdict on any programme or agency.

Bundling. Individual village mini-grids are too small for institutional investment. A typical system serves 200 households at a capital cost of roughly $300,000 — below the minimum ticket for development finance institutions. The bundling layer aggregates villages into investable portfolio units within each kecamatan (sub-district), the smallest administrative boundary where shared logistics, operations, and contractor deployment are physically meaningful. The kabupaten (district) becomes the portfolio vehicle, a collection of kecamatan-level bundles structured as a single Special Purpose Vehicle per the UNEP C2E2 Sourcebook model.

Villages are grouped using Ward-linkage agglomerative hierarchical clustering, a standard method that merges the most similar pair of clusters at each step, stopping when the merge cost exceeds a distance threshold. No minimum or maximum bundle size is imposed; the algorithm finds natural stopping points, and villages too dissimilar to any cluster remain as singletons — a genuine finding, not a flaw. Each village is represented as a point in a 14-dimensional feature space constructed from four weighted groups: technology homogeneity (weight 0.30, one-hot encoded L2 prescription), beneficiary profile (0.30, projected 2026 households, population density, and service quality index), resource similarity (0.25, six z-scored resource features), and intervention type (0.15, binary dark/intermittent classification following Assaf and Assaad 2023). Geographic proximity is handled by the kecamatan scope, not by a feature axis, avoiding the circularity of measuring geographic distance within a geographically constrained search space. Population is projected from the 2017 Podes village census using village-level compound annual growth rates, floored at 0.8% and capped at 5%.

The distance threshold (dendrogram cut height) controls cluster granularity and was selected at 0.55 from a sensitivity analysis sweeping 0.30 to 1.00, at the elbow where multi-village bundle count plateaus and mean coherence begins degrading more steeply. Applied to the 513 tractable off-grid villages across Maluku's 89 kecamatan, the clustering produces 133 bundles (101 multi-village, 32 singletons) with a median size of 3 and a mean Bundle Coherence Score of 0.87. The clustering is validated against four baselines — kabupaten-only grouping, kecamatan-only, technology-only, and random assignment (100 trials) — using silhouette score, Calinski-Harabasz index, Davies-Bouldin index, and within-cluster sum of squares.

Each completed bundle is routed to one of three financing pathways. Bundles with coherence score above 0.55 and at least three villages enter the blended finance track: 79 bundles covering 437 villages. Bundles that fall below this threshold or contain fewer than three villages are routed to the public good track (54 bundles, 76 villages) for concessional or grant funding. Bundles scoring below 0.35 would be deferred for reassessment with new data; none currently fall in this category.

Financial benefit analysis. System sizing follows the ESMAP firm capacity convention. Capital expenditure is built up from components using IRENA 2026 technology cost data.

Revenue is modelled using the Perpres 112/2022 HPT, which specifies technology-specific tariffs in cent USD/kWh multiplied by the location factor F in years 1–10 and a lower flat rate in years 11–20. Systems incorporating battery storage receive an additional revenue component capped at 60% of the generation tariff per Pasal 10. This time-varying, tech-specific tariff structure produces a weighted 20-year average of $0.174/kWh for solar-battery, $0.194/kWh for solar-biomass hybrid, $0.171/kWh for wind-solar hybrid, and $0.108/kWh for micro-hydro. Operating expenditure is estimated at technology-specific rates per IRENA 2026 Annex C: solar PV 2.0%, onshore wind 3.0%, and BESS 2.5% of installed cost per year. Discount rate is 7.5% real per IRENA 2026 Annex D for non-OECD contexts, with sensitivity tested at 5.0% (concessional) and 10.0% (commercial).

The resulting net cash flow stream — year 0 CAPEX offset by 20 years of tariff revenue minus O&M — produces a pipeline-level NPV of $185M at the base discount rate, with mean IRR of 26.4% and mean payback of 5.2 years. The distribution is bimodal: 383 solar, biomass, and wind villages achieve IRR of 20–44%, driven by the Pasal 10 battery adder that explicitly rewards storage-paired generation. The 130 micro-hydro villages are NPV-negative at the regulated PLTA tariff ceiling, which was set nationally and does not

cover the civil works cost of run-of-river systems on remote islands. This finding validates the three-exit routing: micro-hydro villages route to the public good track where the social return — electrifying communities with zero current access — justifies sovereign investment despite negative financial return. Mean LCOE across the pipeline is $0.105/kWh against an average tariff of $0.167/kWh; per-connection cost of $1,995 falls within ESMAP's surveyed range of $1,000–2,500.
All financial figures are indicative and subject to field validation.

Environmental benefit analysis. The counterfactual for each off-grid village is a diesel generator operating 8 hours per day at 33% thermal efficiency — the standard configuration for rural PLTD systems across eastern Indonesia. Lifecycle greenhouse gas intensity for this counterfactual is 900 gCO_2eq/kWh of delivered electricity, comprising 808 gCO_2eq/kWh from direct combustion (IPCC Emission Factor Database, 266.76 gCO_2/kWh of fuel energy divided by 0.33 generator efficiency) plus approximately 15% for upstream extraction, refining, and transport (UNECE 2022; IPCC AR5 Annex III, Schlömer et al. 2014). The midpoint of the published range (650–1,100 gCO_2eq/kWh) is used throughout.

Each renewable technology carries its own lifecycle emission factor drawn from the same sources: micro-hydro (run-of-river) 24 gCO_2eq/kWh, solar PV 27 gCO_2eq/kWh, wind-solar hybrid 15 gCO_2eq/kWh (weighted 60% onshore wind at 11, 40% solar PV at 27), and solar-biomass hybrid 88 gCO_2eq/kWh (weighted 70% solar PV at 27, 30% dedicated solid biomass at 230). The biomass midpoint assumes sustainably sourced feedstock without land-use-change emissions; biomass linked to deforestation can exceed diesel's lifecycle intensity, so this assumption must be validated at project development stage. Net avoided emissions per village equal annual generation (demand in kW × 8 hours × 365 days) multiplied by the difference between the diesel and RE lifecycle emission factors, divided by 10^6 to convert grams to tonnes.

Applied to the 513-village pipeline, total avoided emissions are approximately 230,000 tCO_2eq per year, or 5.8 million tCO_2eq over the 25-year asset life. The calculation is lifecycle rather than combustion-only: it credits the full upstream diesel supply chain displaced while debiting the manufacturing, construction, and decommissioning footprint embedded in each RE technology. This net accounting is more conservative than the combustion-only approach (which would use 700 gCO_2/kWh per IPCC AR6 and ignore RE embodied emissions) but more defensible.

Socioeconomic benefit analysis. Direct measurement of household-level economic and social benefits requires field survey data that does not yet exist for Maluku's off-grid pipeline. Instead, benefits are estimated by transferring a benefit-cost ratio from Wibisono, Lovett, Wen, and Suryani (2023), who conducted a cost-benefit analysis of the Kalilang micro-hydro plant on Sumba Island — a community-managed off-grid system serving 280 rural households in a context closely comparable to the Maluku pipeline: eastern Indonesia, subsistence agriculture, kerosene-baseline lighting, and externally funded RE infrastructure handed to community management.

Wibisono et al. identify two market benefits — kerosene expenditure substitution and productive-use income uplift from extended working hours (mat weaving under electric light) — and five non-market benefits valued through double-bounded dichotomous choice contingent valuation (entertainment, information access, clean water pumping, study lighting, and customary ceremony convenience). Their Table 5 reports a benefit-cost ratio of 3.43 for market benefits inclusive of installation cost over a 50-year plant life at a 5.75% discount rate. Only the market-benefit BCR is transferred; the non-market component (which would raise the ratio to 11.32) is excluded because it derives from site-specific willingness-to-pay estimates that cannot be assumed transferable without a Maluku-specific survey. This makes the estimate conservative.

The transferred BCR is applied to annualised project cost: total CAPEX multiplied by a capital recovery factor plus annual operating expenditure. The capital recovery factor is calculated at a 9% social discount rate — the minimum required economic internal rate of return for developing member countries per ADB (2017), *Guidelines for the Economic Analysis of Projects* — over a 25-year asset life per IRENA's standard economic life assumption for solar and wind assets. The 9% social discount rate is distinct from the 7.5% financial discount rate used in the financial benefit analysis: the financial rate reflects what private investors require to commit capital, while the social rate reflects the opportunity cost of public resources and is the standard benchmark for economic appraisal of development projects across ADB member countries.

At these parameters (CRF = 0.1018), the pipeline generates an estimated socioeconomic benefit of approximately $108 million per year, or $844 per household per year — representing the annualised value of kerosene savings and productive-use income gains that electrification is expected to deliver, based on measured outcomes in a comparable Indonesian off-grid context. The benefit-to-investment ratio over 25 years is approximately 10.6×.

Exit routing. Each completed bundle is evaluated on two dimensions: financial viability (NPV and IRR from the component-based financial analysis) and bundle coherence (the four-axis BCS computed post-hoc from the clustering). Bundles are routed to one of three financing pathways:

1. **Blended finance pathway.** The bundle achieves positive NPV at the base discount rate and a coherence score above 0.55 with at least three villages. It proceeds to the district portfolio vehicle for blended capital deployment (concessional, commercial, guarantee). In the Maluku pilot, 79 bundles covering 437 villages and 383,000 beneficiaries enter this pathway, with a mean IRR of 27.8 per cent.
2. **Public good pathway.** The bundle demonstrates clear social impact — electrifying communities with zero or degraded access — but cannot generate sufficient economic return at the regulated tariff ceiling. It is routed to purely concessional or grant-funded instruments, framed as a public good investment justified on development grounds. The 130 micro-hydro villages fall here: the nationally-set PLTA tariff does not cover the civil works cost of run-of-river systems on remote islands, but the social case for electrification is unambiguous. Facilities such as the ACGF and MDB climate funds hold mandates specifically designed for this class of investment.
3. **Deferred pathway.** The bundle clears neither financial viability nor minimum coherence thresholds. It is parked and flagged for reassessment when new data — updated population counts, ground-truth demand surveys, or regulatory changes to the HPT schedule — alters the input parameters. Villages in a deferred bundle remain in the pipeline; their prescriptions are not withdrawn. No bundles in the current Maluku pilot fall in this category.

The routing is deterministic: given a fixed bundle composition and tariff regime, the exit pathway is fully determined. The three-exit structure generates an empirical signal about the gap between technical viability and financial viability — the bimodal IRR distribution, with a dead zone between 5 and 18 per cent where no bundles sit, directly informs the appropriate ratio of concessional to commercial capital. The design draws on established practice in regional energy planning where diversified portfolios outperform single-pathway strategies (Anugrah et al., 2026), and on Indonesia's own tariff framework which recognises that electricity delivery costs in Maluku carry a 25–30 per cent location premium over Java (Perpres 112/2022; Midadan, INDEF, 2023).

Future work may introduce iterative re-assembly, in which bundles that fail financial screening are re-composed — widening the geographic boundary to an adjacent kecamatan or shifting the technology mix — and re-evaluated. The current implementation treats re-assembly as a manual process informed by the dashboard's bundle catalog rather than an automated loop, because the three-exit routing already directs every village to an appropriate capital pathway without requiring re-composition.

Key Components/Stakeholders

Development financiers. Facilities such as the ASEAN Catalytic Green Finance Facility, the ASEAN Infrastructure Fund and multilateral development banks hold capital mandated for energy access but face a shortage of bankable projects. They are the primary customer, receiving a pre-screened pipeline and a standardised monitoring layer that lowers due-diligence cost per project.

Government and utilities. ESDM, PLN and district governments gain an independent prioritisation instrument for final-mile programmes. The framing matters: Terang helps government reach its own universal access target sooner, rather than auditing past claims.

Mini-grid developers. Qualified leads with indicative technology and sizing, plus faster milestone confirmation, easing the cash-flow constraint verification delays impose today.

Communities and village energy committees. Residents are not passive beneficiaries. Satellite flags are hypotheses; community members confirm or reject them through a simple mobile form and are paid per verification, and elected committees co-sign milestone confirmations. This resolves the method's main technical weakness, creates local ownership, and generates ground-truth data that improves the system over time.

Researchers and civil society. The methodology, including its limitations and the masking correction above, is published openly so it can be reproduced and extended.

Impact and Sustainability

Impact potential. As shown in Figure 9, in Maluku, 111 villages appear settled yet effectively dark, concentrated in Aru Islands (26), Buru Selatan (19), Seram Bagian Timur (15) and Tanimbar Islands (13), with the majority of remaining villages showing unstable service. Nationally, government figures identify 4,808 villages without a PLN connection. The method should transfer to any jurisdiction for which a village boundary file exists.

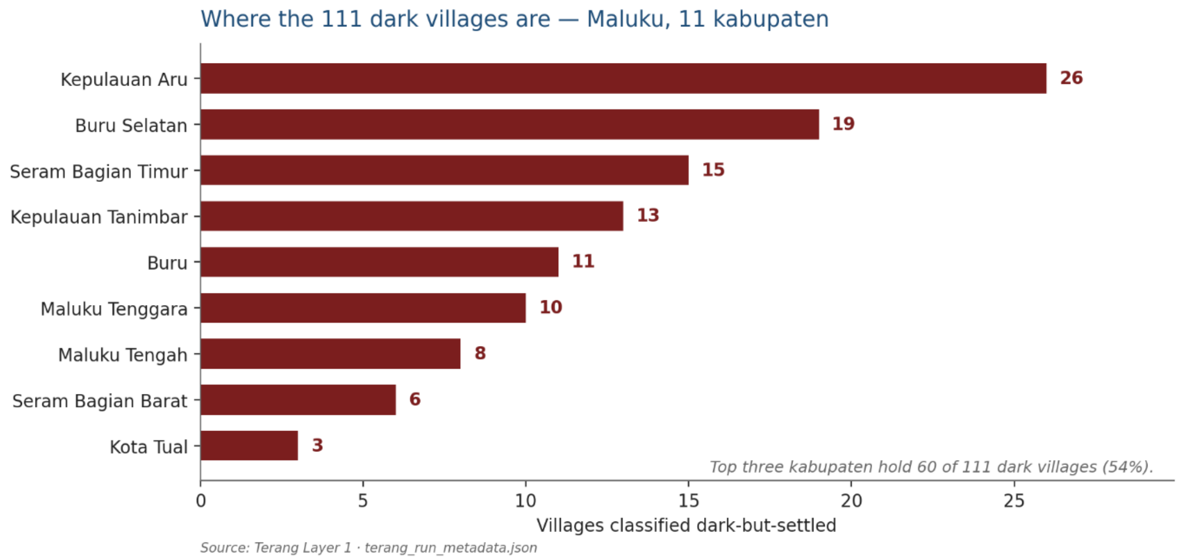

Figure 9. Dark-but-settled villages by Regency: Aru Islands 26, Buru Selatan 19, Seram Bagian Timur 15, Tanimbar Islands 13, Buru 11, Maluku Tenggara 10, Maluku Tengah 8, Seram Bagian Barat 6, Kota Tual 3.

Socioeconomic and environmental impact. Renewable energy projects in remote archipelagic settings are rarely justified by financial returns alone. Terang addresses this by computing an SCBA and environmental benefit for each village cluster (Table 1), translating satellite-observed electrification into monetised and non-monetised impact across three dimensions. Economically, reliable electricity enables productive-use applications – cold storage, fish-drying, milling, ice-making – that raise household incomes and extend market reach beyond what lighting alone achieves; local construction and operation of mini-grid systems also generate direct employment. From a welfare standpoint, replacement of kerosene and diesel systems removes a fuel-cost burden that can represent ten to twenty per cent of household expenditure in off-grid communities (Bacon, Bhattacharya & Kojima, 2010; BPS Susenas 2023), while extended evening hours support education and health outcomes observable over multi-year radiance records. Environmentally, the conservation screen built into Layer 2 diverts projects away from protected and intact forest — each village prescription carries an implied figure for avoided infrastructure encroachment; and where solar-battery or micro-hydro systems displace existing diesel generators, the displaced fuel volume translates to a calculable avoided-emission estimate per village per year. The goal of this project is therefore not simply to increase the number of villages counted as electrified, but to improve the reliability and quality of electrification service — and to demonstrate that doing so produces measurable societal returns that justify the concessional share of blended-finance structures.

SCBA and environmental impact matrix - illustrative per village class

Benefit dimension	Dark settled (111)	Intermittent (894)	Existing PLTD (diesel)	Metric / source
Employment	New: construction (2–4 jobs/village), O&M (1 technician), paid verifiers	Retrofit crew + O&M; paid local verifiers at each milestone	Repurposing crew (diesel-to-hybrid); retained O&M roles	FTE per village; ILO job-year coefficients
Household welfare	Full kerosene/diesel saving (~10–20% HH income); productive-use income uplift (cold storage, fish-drying); extended study hours	Partial saving from reduced outages; reliability premium on productive equipment uptime	Reduced PLN diesel subsidy transfer to village tariff; lower household energy cost	WorldPop HH count × unit saving; productive-use revenue survey
Environmental	Forest protected: 24 villages screened to policy pathway = area not encroached; avoided CO_2 vs diesel counterfactual per prescribed RE system	Avoided CO_2 from replacing partial diesel hours; deforestation pressure reduced where biomass feedstock is agricultural residue	Largest avoided-emission potential: full diesel displacement; PLTD-to-PLTG asset conversion reduces imported fuel dependency	tCO_2/yr × carbon price; WDPA + Hansen forest area; Layer 2 capex × emission factor
Electrification quality / reliability	Baseline = 0; target: Service Quality Index ≥ 0.8 (≥80%	Target: SQI uplift from observed baseline to	Target: SQI ≥ 0.9 (hybrid system provides	VIIRS monthly radiance time series; lit_fraction

	months lit above threshold) sustained over 12 months post-commissioning	≥0.8; monsoon-season drop eliminated	higher base reliability than diesel-only)	and rad_cv from Layer 1 pipeline

Table 1. Terang SCBA and environmental impact matrix. SQI = Service Quality Index (fraction of observation months where village radiance exceeds detection threshold). Values are indicative pending population join (WorldPop) and ground-truthing. PLTD = diesel power plant; PLTG = gas-fired equivalent; CO_2 basis: direct combustion EF 0.7 kgCO_2/kWh for diesel reciprocating engines (Clarke, Wei et al., 2022). Note: Annex III reports median lifecycle GHG ~820 gCO_2eq/kWh; we use the direct combustion factor). Employment multipliers: IRENA & ILO, Renewable Energy and Jobs: Annual Review, 2024. ILLUSTRATIVE fields (Figure 10 verification cost comparison): no published field-audit benchmark currently available for Maluku; figures are indicative pending formal quote from an M&E firm operating in eastern Indonesia.

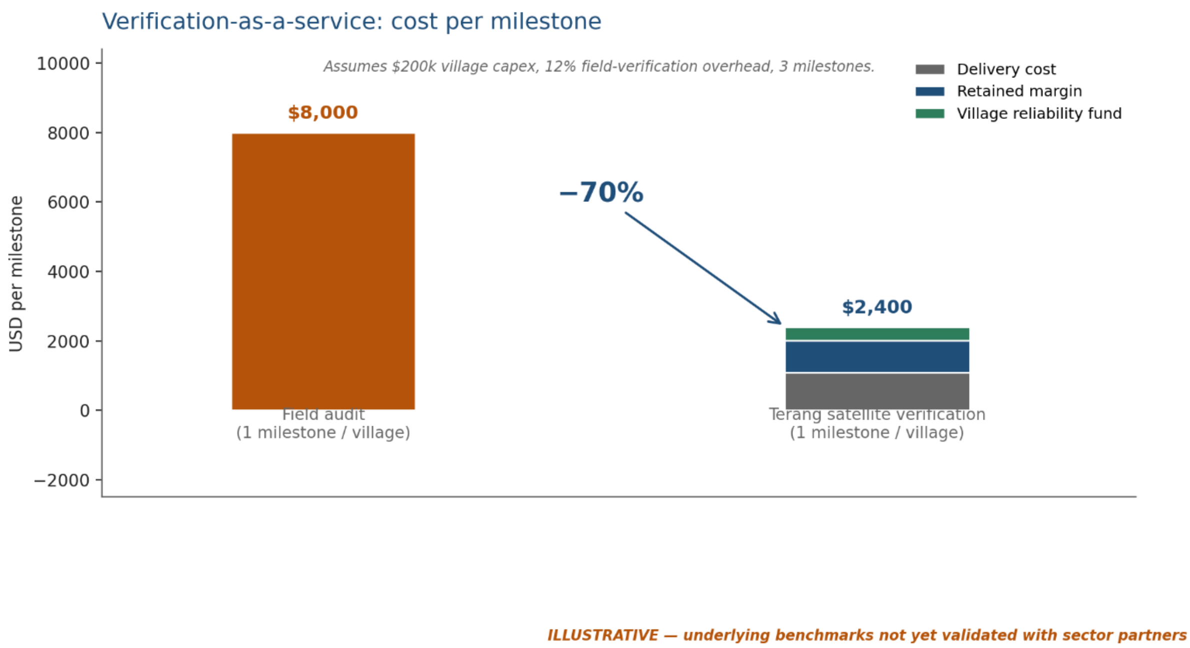

***Figure 10.** Unit economics of the verification-as-a-service model described under Future Possibilities: cost of conventional field verification per village per milestone against the Terang satellite-verified equivalent, with the resulting margin and the ring-fenced share directed to the village reliability fund.*

Two product lines. The dark villages are the accessibility case: new capacity, clear additionality, strong concessional appeal. The villages with unstable service are the commercial case: retrofit of failing systems, lower cost per unit, faster deployment, larger pipeline. The first proves the mechanism on the hardest cases; the second is where volume and return reside. Financial feasibility for the 73-village dark pipeline is modelled in Appendix I.

Business model. Terang is asset-light, with three revenue lines: origination fees for a qualified pipeline; verification-as-a-service charged per milestone, priced below the field audit it replaces; and data subscriptions for planners and utilities. Because the data is free and the analysis automated, marginal cost per additional village approaches zero, while the accumulating archive of ground-truth confirmations becomes an asset competitors cannot easily replicate. Verification also identifies villages where energy is used productively – for cold storage, ice-making, fish-drying, coconut processing or milling – rather than for lighting alone. Because productive use raises household earnings and therefore willingness and ability to pay, Terang treats it as a leading indicator of persistence and preferentially routes concessional capital to developers who bundle productive equipment with generation.

Social enterprise structure. To prevent drift toward serving financiers at the expense of communities, a fixed share of verification revenue is ring-fenced into a village reliability fund that pre-finances the repairs monitoring detects are needed. The commercial engine funds the social outcome rather than competing with it.

Sustainability. The model does not depend on grant funding: inputs are free, costs scale sub-linearly with coverage, and customers hold standing budgets for these functions. By screening sites against protected areas and intact forest before recommending them, it also avoids the conflict between energy expansion and conservation that has constrained renewable deployment across Indonesia.

Timeline

Phase	Description	Duration
Pilot Validation	Complete detection and resource screening for Maluku; add population, official electrification records and conservation layers	3 months

Ground-Truth & Community Pilot	Field-confirm flagged villages; recruit and train paid local verifiers; establish village energy committees	4 months
First Close – Aru Islands	Fund the 16-village demonstration tranche; select developers; agree the milestone schedule with financiers	6 months
Provincial Scale-Up	Extend to the full 64-village Maluku portfolio and open the retrofit line for villages with unstable service	12 months
National Replication	Deploy to Papua and Nusa Tenggara, where the recorded access gap is largest; integrate with national programmes	18 months
Regional Expansion	Adapt to the Philippines and other ASEAN archipelagic states; license verification to regional green finance facilities	24 months+

Conclusion

Terang is a system that uses satellite imagery to find villages that don't have reliable electricity – even when official records say they do – and translates that finding into screened, bundled investment-grade project pipelines. Rather than treating electrification as a binary connection event, Terang treats it as a measurable, continuous condition: identifying which villages are dark, prescribing what should be built in each, bundling them into portfolios large enough for institutional capital, and validating each bundle through socioeconomic and environmental benefit analysis before it reaches any financier.

On accessibility, Terang finds villages that official records miss – 111 in Maluku alone, recovered only after correcting a standard geospatial method that was systematically discarding the smallest and most remote settlements. On energy security, it directs villages toward local renewable generation rather than shipped-in diesel, reducing fuel-price vulnerability and storm exposure. Its prescriptions are cross-validated at 98% against Kementerian ESDM's official resource registry – an independent confirmation that no comparable satellite-based system has previously demonstrated at this coverage. And its pooling mechanism transforms individually unbankable village projects into district portfolios where the investment case is grounded in quantified social returns, not assumed.

The verification and milestone-gated disbursement mechanism – where developers are paid only when satellites and community verifiers confirm that light persists – is described under Future Possibilities as the next operational layer. The core contribution of this proposal is the pipeline itself: cheap, continuous, independent observation that makes the missing villages visible and investable for the first time.

Future Possibilities

The three-layer pipeline described above – Detect, Prescribe, Pool – delivers a screened, validated, and bundled project pipeline ready for capital deployment. The following capabilities represent the natural next stage: mechanisms for governing how capital flows from portfolio vehicles to developers, and how ongoing performance is incentivised after commissioning. These are design-ready extensions, not speculative features; the satellite infrastructure and data architecture required to support them are already built into the pipeline.

Milestone-gated disbursement. Once a district portfolio vehicle is capitalised, disbursement to developers can be tied to satellite-verified events rather than field attestations. Commissioning (M1) would be detected as a step change in nighttime radiance – the same function that identified Labuang village's January 2017 electrification from data alone. From that point, two independent monitoring clocks run concurrently: persistence (M2) re-confirmed on a rolling three-month basis, and sustained service (M3) on a rolling twelve-month basis. Each satisfied window releases its own disbursement tranche, checked in parallel rather than sequentially. This design ensures that developers are paid for light that persists, not merely for connections installed – directly addressing the persistence failure identified in the problem statement.

Verification-as-a-service. Every M1, M2, and M3 confirmation generates a fee to Terang's verification bureau, distinct from the capital moving to the developer. A satellite flag is a hypothesis, not a fact, until a paid local resident confirms it through a mobile form and an elected village energy committee co-signs the milestone – this is the only way persistence can be checked monthly rather than once, at a fraction of the cost of a conventional field audit that is typically performed a single time and never repeated.

Village reliability fund. A fixed, ring-fenced share of every verification fee flows automatically into a community-managed fund that finances repair or replacement directly when ongoing monitoring detects a village has gone dark again – closing the loop between detecting a failure and funding its fix. The commercial engine funds the social outcome rather than competing with it.

Productive-use modifier. A modifier applies where satellite radiance patterns are consistent with productive load – sustained draw during pre-dawn or post-sunset hours, of the kind produced by cold storage, ice-making, drying racks, or milling equipment rather than by lighting alone. Villages exhibiting this signature would release an additional persistence tranche to their developer, on the reasoning that productive use raises household earnings and therefore willingness and ability to pay. In Kepulauan Aru specifically, a solar-battery system paired with a shared cold room extends catch shelf life from hours to days, reaching buyers a lighting-only system never could – the same radiance signal from space, a materially different outcome on the ground.

Deeper Secondary data integration and cross-validation. The current pipeline cross-validates prescriptions against Indonesian Ministry of Energy and Mineral Resource (ESDM) registry at 98% agreement. Future iterations will incorporate ESDM's conventional infrastructure data – existing PLTD (diesel) locations, substation distances, and transmission corridors – to evaluate asset repurposing and hybrid configurations such as solar-diesel transitions. The 13 "ghost generator" villages identified in this pilot (ESDM-registered off-grid systems built 2012–2016 that no longer produce detectable radiance) represent a particularly valuable dataset for understanding system degradation patterns and calibrating maintenance intervention timing.

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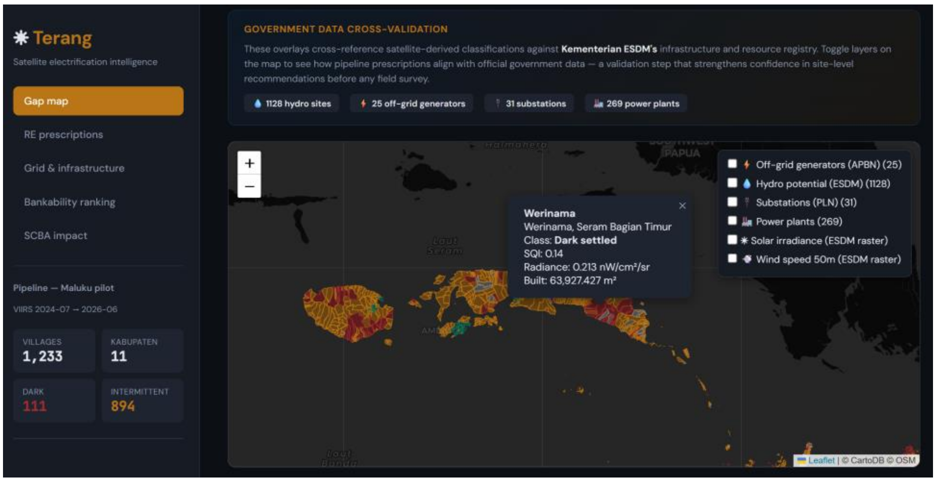
Appendices

Appendix - Mock Up

The demonstration application presents the pipeline in three linked views, built on real Maluku data. View 1, Gap Map: a province-wide map with every village coloured by class – dark but settled, intermittent, stable or unresolved – where selecting a village shows the official claim against the observed light record and a 22-month history. View 2, Prescriptions: the same map recoloured by recommended technology, each card stating technology, indicative capacity, cost band and the decision rule that produced it, so no output is a black box. View 3, Portfolio and Milestones: district bundles with aggregate cost and technology mix, a milestone board per village, and the verification chart showing monthly radiance against the detection threshold with the detected commissioning step.



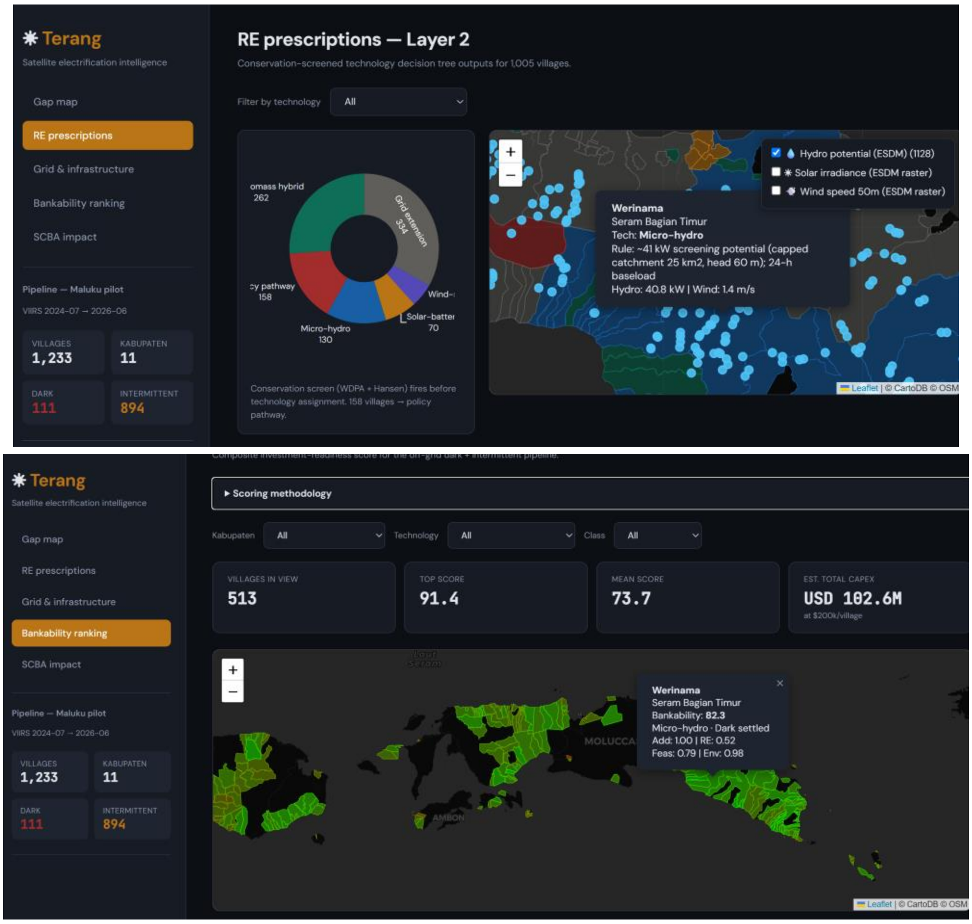


FIGURE A1. Demonstration application, (i) View 1 Gap Map at province zoom; (ii) View 1 with a single dark village selected, showing its 22-month radiance history; (iii) View 2 Prescriptions with a recommendation card and its decision rule visible; (iv) View 3 Portfolio and Milestones showing a district bundle and the verification chart.

Appendix - Working Flowchart

The pipeline runs in two stages. In Stage A, village boundaries, satellite radiance and settlement data are combined in a cloud geospatial platform: statistics are computed per village, the threshold calibrated from uninhabited reference points, villages classified, and flagged villages passed through resource and conservation screening. In Stage B, the application reads those exports and presents the three views. Milestones are evaluated by the same functions used in analysis, so the logic shown to users is identical to the logic used to classify. Because analysis runs once against the archives and the application reads only exported results, it stays fast and works without network access.

Flow: village boundaries and satellite imagery → radiance statistics and threshold calibration → classification → resource and conservation screening → technology prescription → portfolio bundling and SCBA validation. The downstream milestone verification and disbursement stages are described under Future Possibilities.

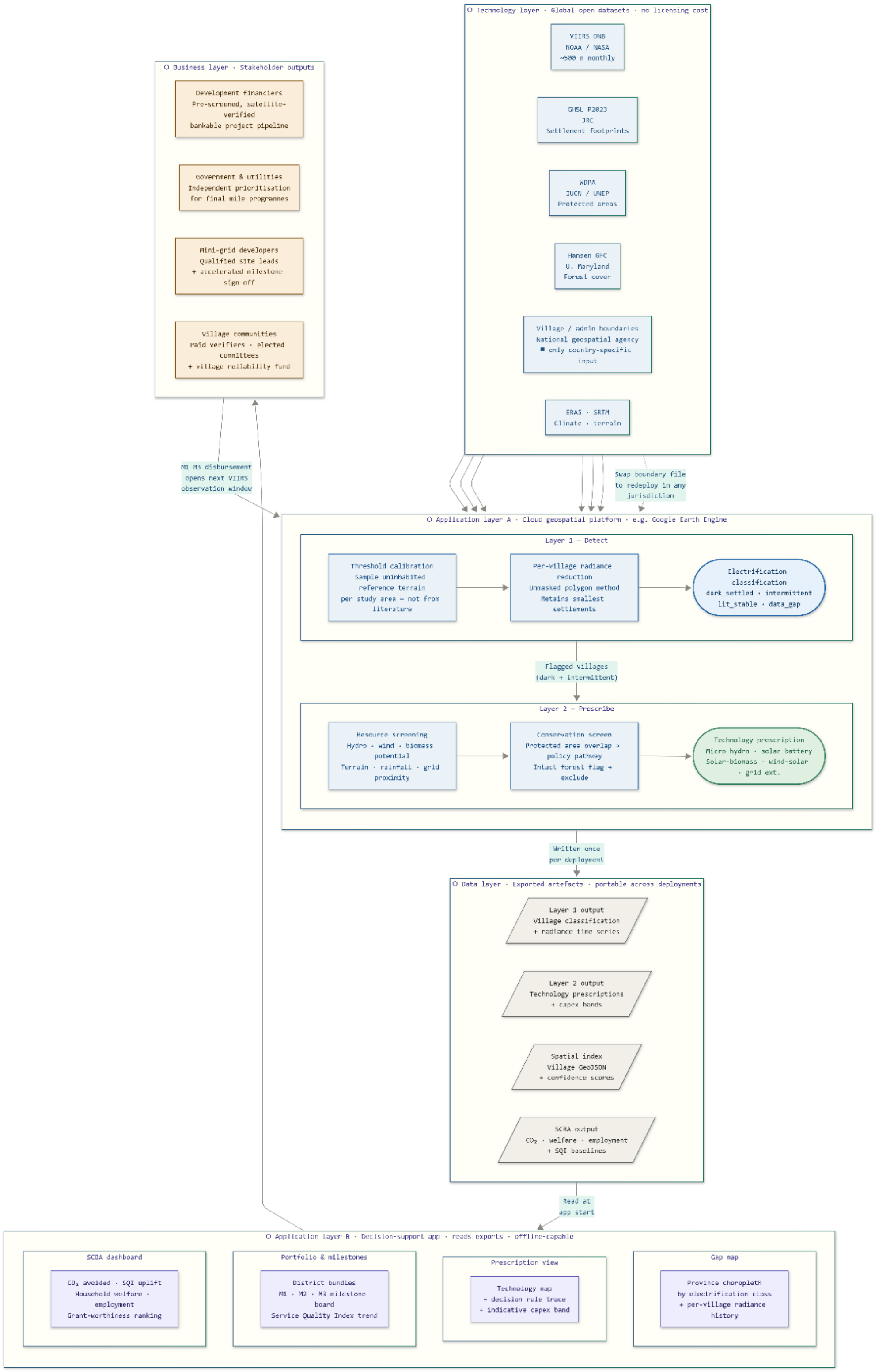


FIGURE A2. End-to-end pipeline flowchart. Stage A in the cloud geospatial platform: village boundaries and satellite imagery, radiance statistics and threshold calibration, classification, resource and conservation screening, technology prescription. Stage B in the application: portfolio bundling and SCBA validation. The milestone verification and disbursement stages shown in the lower portion represent the Future Possibilities extension described in the main text.

Team Composition

Kartiko Cokro Sewoyo – technical lead. GIS Analysis Team Leader on the Budget Execution Data Analytics task force at Indonesia's Ministry of Finance, where he built a satellite change-detection system verifying whether government-funded agricultural programmes produced real outcomes on the ground. MSc in Data Science, University of Manchester. Responsible for the detection pipeline, resource screening, milestone verification logic and the demonstration application.

Meiliza Fitri – energy and financing lead. State Asset Analyst at the Directorate General of State Assets Management, Ministry of Finance, with experience in energy asset valuation, policy research, and state asset optimization, including coal mining and oil and gas infrastructure assets. M.Eng. in Energy and Resources Engineering from Chonnam National University, South Korea, and former Lecturer in Mining Engineering at Jambi University. Author and presenter of multiple international publications on energy transition, CBM reservoirs, EOR, LNG infrastructure repurposing, and state asset policy. Responsible for energy technology benchmarks, integration of energy models, financing structures, and regulatory policy framing.

Dwiky Anggawan Pradiptya Putra – policy and institutional lead. Policy Analyst at the Directorate General of State Assets Management, Ministry of Finance, with prior work on state asset revaluation, the simplification of five ministerial regulations into a single framework (PMK 173/PMK.06/2020), and analysis of General Allocation Fund spending patterns for the Directorate General of Fiscal Balance. Master of Public Policy, University of Michigan (Ford School). Responsible for policy pathway design, regulatory alignment with PLN and ESDM frameworks, and the institutional adoption route into Indonesian public finance.

The team combines geospatial and data science capability, energy sector and asset optimisation expertise, and direct policy access within Indonesian public finance – with a working relationship across three directorates general of the Ministry of Finance. This provides both a route to primary data and a credible pathway to institutional adoption. Team roles and responsibilities have been divided to allow parallel work streams; all three members are committed to attending the Manila finals if selected.

References

Azura Yumna Ramadani Purnama. (2026). Bloomberg Technoz: ESDM Anggarkan Listrik Desa 2026 Rp10,3 T, 1.520 Lokasi Kontrak. 5 June 2026. https://www.bloombergtechnoz.com/detail-news/110994/esdm-anggarkan-listrik-desa-2026-rp10-3-t-1-520-lokasi-kontrak

Bacon, R., Bhattacharya, S., & Kojima, M. (2010). Expenditure of Low-Income Households on Energy: Evidence from Africa and Asia. World Bank Extractive Industries for Development Series No. 15.

Bedah Nusantara. (2025). Masih Ada 130 Desa di Maluku Belum Terlistriki PLN. 8 November 2025. https://bedahnusantara.com/masih-ada-130-desa-di-maluku-belum-terlistriki-pln/11/08/

BPS (Badan Pusat Statistik). (2023). Susenas: Survey Sosial Ekonomi Nasional. Jakarta: BPS Indonesia.

Center for Strategic and International Studies. (2026). The Indispensability of ASEAN China Relations: Retrospect and Outlook, in cooperation with China Foreign Affairs University

Clarke, L., Wei, Y.-M., et al. (2022). Energy Systems. In: Climate Change 2022: Mitigation of Climate Change (IPCC AR6 WG3, Chapter 6 + Annex III Table AIII.2). DOI: 10.1017/9781009157926.008.

Covenant of Mayors for Climate & Energy Office. (2023). A “Project Bundling” approach for implementing climate/energy related interventions - Bundling projects from different local authorities with similar contexts and within specific thematic areas. August 2023

Eco-Business. “ASEAN Power Grid Needs at Least 100 Billion US Dollars to Build Transmission Lines by 2045: ADB.” Accessed October 8, 2025. https://www.eco-business.com/news/asean-power-gridneeds-at-least-us100-billion-tobuild-transmission-lines-adb/.

Ember. Wired for Profit: How Power Companies in ASEAN Are Cashing In on the Energy Transition. London: Ember, 2025. https://ember-energy.org/app/uploads/2025/05/ReportWired-for-profit-ASEAN-PDF.pdf.

ESDM. (2024). ESDM One Map - Exploring Energy and Mineral Resources of Indonesia. https://geoportal.esdm.go.id/home/

ESMAP. (2018). Small Hydropower Development Planning Tool. World Bank, Washington DC.

ESMAP. (2019). Mini Grids for Half a Billion People: Market Outlook and Handbook for Decision Makers. World Bank Report No. 132039.

ESMAP. (2020). Geospatial Planning Tools for Universal Electrification. World Bank, Washington DC.

IEA. (2021). Hydropower Special Market Report. International Energy Agency, Paris.

IRENA. (2019). Future of Wind: Deployment, Investment, Technology, Grid Integration and Socio-Economic Aspects. ISBN: 978-92-9260-155-3.
IRENA & ILO. (2024). Renewable Energy and Jobs: Annual Review 2024 (11th edition). https://www.ilo.org/publications/renewable-energy-and-jobs-annual-review-2024
Kementerian ESDM. (2023). Statistik Ketenagalistrikan. Direktorat Jenderal Ketenagalistrikan, Jakarta. https://gatrik.esdm.go.id/
Mentis, D., et al. (2017). Lighting the world: the first application of an open source, spatial electrification tool (OnSSET) on Sub-Saharan Africa. Environmental Research Letters, 12(8), 085003. DOI: 10.1088/1748-9326/aa7b29
Min, B., O'Keeffe, Z.P., et al. (2024). Lost in the dark: A survey of energy poverty from space. Joule, 8(6). DOI: 10.1016/j.joule.2024.05.001
Anugrah, P., Shah, R., Mithulananthan, N., et al. (2026). Sustainable energy transition for BIMP regional power grid with technology ranking and optimal mix. Applied Energy, 424, 128463. DOI: 10.1016/j.apenergy.2026.128463.
Midadan, M. (2023). Analisis Tekno-Ekonomi Pengembangan Panas Bumi dan Energi Terbarukan Indonesia. INDEF, Jakarta.
Perpres (Peraturan Presiden) Nomor 112 Tahun 2022 tentang Percepatan Pengembangan Energi Terbarukan untuk Penyediaan Tenaga Listrik. Republik Indonesia.
PLN. (2023). PLN Annual Report 2023. PT Perusahaan Listrik Negara, Jakarta.
World Bank / DTU Wind Energy. (2021). Global Wind Atlas 3.0. https://globalwindatlas.info